

EXP1-4 Self-powered vehicle Report P14

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Section1-Introduction :

For this experiment, the objective is to build a self-powered vehicle. The main problem is the contradiction between demand and real situation. The vehicle needs a smooth process of energy conversion. However, there is a lack of materials in the real situation. To solve the problem, some key factors are proposed: stability, route straightness, cost-efficiency, and sustainability. By following key factors, a gravitational potential energy self-powered vehicle was successfully fabricated.

Section2-Design Strategy :

Our group came up with a strategy, which *transferred gravitational potential energy to kinetic energy* to move the car forward. And the vehicle can be seen as two parts: *energy storage device* and *base*.

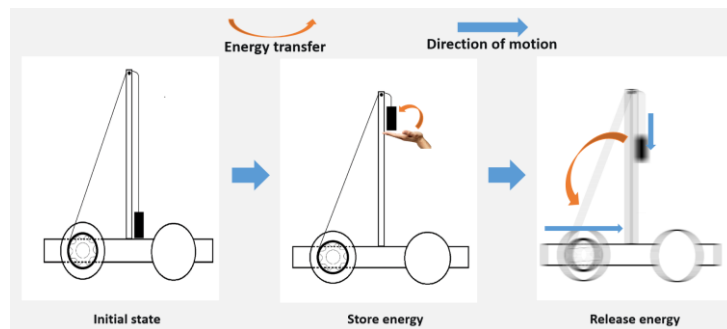


Fig.1 the principle of energy storage device

For *energy storage device* shown in Fig.1, when pulling the weight up to the highest point from the initial state, the device stored gravitational potential energy. Then, as the weight descends, the rope pulls the axle of the rear wheel forward. The process accomplished the transfer of gravitational potential energy to kinetic energy.

To achieve the transformation of energy, our group first put the axle of the rear wheel through the pulley made of gears and fixed the pulley with hot glue. We used boxes of scrap tape as the main body of the car and then tape several chopsticks together and secure them to the car. The top of the bracket is connected by a smooth thin rod. Finally, we wrap the tape around the pulley enough times to wrap the other end around a thin rod and attach a weight. This structure is an important part of the energy conversion function.

For the driving mechanism, when the weight is released from a high place, the force of gravity causes the weight to fall naturally, while the weight uses the tape to turn the pulley, which turns the power wheel (Fig.2 Gravity and Pulling force). And the sliding friction between wheels and ground will push the vehicle forward (Fig.2 Sliding friction). In this process, the stored gravitational potential energy of the weight is converted into the overall kinetic energy of the car moving forward.

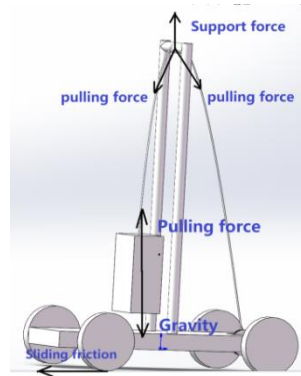


Fig.2. Force analysis of vehicle

Structure	Advantages	Disadvantages
Gears and weights connected by tape (Energy conversion device)	More stable than a car that uses rubber bands to store energy	More space occupancy
A stand made of chopsticks and duct tape	Lighter than the iron, allowing the car to travel further	Multiple chopstick connections may be unstable
The wheel radius that provides power is large	With the same number of turns of the shaft, the larger wheel rolls more distance than the smaller wheel	Heavier and may hard to rotate due to large radius

Tab.1. Advantages and Disadvantages of vehicle

To obtain the relationship between the forward distance of the vehicle and the stored energy, we listed the following formula based on ***the law of conservation of kinetic energy***:

$$\begin{aligned} \eta m_1 g h &= f s \\ f &= \mu m_2 g \end{aligned} \quad \Rightarrow \quad s = \frac{\eta m_1 h}{\mu m_2}$$

where

m_1 is the weight of hanging object

m_2 is the weight of the vehicle (without hanging object)

η is the mechanical efficiency of machine

μ is the dynamic friction factors of the vehicle

Considering the friction between tape, iron bar and axle, we introduce the mechanical efficiency of pulley block operation η and sliding friction factors μ to ensure the accuracy of the calculation.

Section3- Design cost:

Material	Amount	Total Price
Ice cream sticks	4	1
Recycled steel rods	2	0.5

Tab.2. Design cost of vehicle

The total cost of our group is 1.5 QMES \$

Section4- Sustainability:

Based on the recycling, reusing, and reducing strategies, our team used the wheel, toy car gears, pirate cassette, repurposed chopsticks, and old batteries in the experiment. These recycled materials were used to build the vehicle together and **everything can be recycled. The weight of vehicle is 80g.**



Fig.3.Material that dissambled from vehicle (All have been returned to QMES stores)

Our team made the *optimum use of gravitational potential energy transformed to kinetic energy for long-term energy sustainability*. Gravitational potential energy, which can be stored multiple times exactly, transferred into kinetic energy without other energy loss, while elastic energy in the rubber band will decrease with each use.

To lower the friction between tape, iron bar and axle, our team decided to *substitute the cassette tape for the string in order to reduce friction*. Due to its low coefficient of kinetic friction, this not only minimizes friction but also adheres to the reducing principle.

Section5- Prediction and Performance Evaluation

In the test, *most of the parameters were in line with our predictions*, with the only difference being that the vehicle traveled a 2.65m instead of 3.36m, **the error is 21%**. Because of the simplicity of the structure, our energy calculation accuracy is almost 100%, some of the energy is lost and not used for vehicle forwarding.

We analyzed this result and speculated that the main reason was that we *omitted some resistance in our calculations*, such as friction between individual moving parts. For these parts, we also think of some solutions, such as adding some lubricant in some places where the friction is strong. For the vehicle race, our target distance is 6.7m, it is difficult to achieve for our design because we need to increase the distance traveled by heavy objects or increase the mass of heavy objects to get more gravitational potential energy. However, this can easily lead to skew of the body or material failure.

	Prediction	Actual
Describe energy storage mechanism	Gravitational potential energy is converted to kinetic energy	Gravitational potential energy is converted to kinetic energy
Total Weight Vehicle	80g	80g
Amount of energy stored	0.98J	0.98J
Energy needed to achieved your performance targets	1.96J	0.98J
The height at which the weight is released	0.45m	0.45m
Design Cost	1.5 QMES dollars	1.5 QMES dollars
Total Distance travelled	3.36m	2.65m
Sustainability credentials	Plastic boxes, rubber bands and chopsticks can be recycled	Plastic boxes, rubber bands and chopsticks can be recycled

Tab.3. Prediction and Actual result